

Reduction in Time-to-Diagnosis for Lung Cancer Resulting From Implementation of a Formal Incidental Pulmonary Nodule (IPN) Program Compared With Traditional Referral Pathways

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Background: Incidental pulmonary nodules (IPN) are nodules identified on imaging performed for a reason unrelated to the diagnosis of cancer. This study analyzes improvement in time-to-diagnosis for patients first presenting with an IPN that was later diagnosed as lung cancer through a formal IPN program.

Methods: This is a 5-year retrospective, single-center analysis of outcomes for IPN patients diagnosed with lung cancer who presented through a formal IPN program (FIP) pathway compared with a matched group of patients diagnosed with lung cancer presenting through traditional IPN referral (TIR) pathway. The primary endpoint was time-to-diagnosis of lung cancer. A chart review was performed to determine the date of the initial CT scan identifying the IPN to the date of the procedure to diagnose lung cancer in the FIP group and the TIR group. The number of days from identification of the IPN to diagnosis was compared between the 2 groups.

Results: Between May 2019 and May 2024, a total of 9303 patients were identified with IPNs through the formal IPN program. Ninety-six of these patients (1%) were diagnosed with cancer following nodule biopsy. Eighty-one of the patients (84%), making up the FIP

group, were diagnosed with primary lung cancer. Twenty-one of the 81 primary lung cancer patients in the FIP group (27%) were diagnosed with early-stage disease and underwent curative resection. The average time from identification of an IPN to diagnosis of lung cancer in the FIP cohort was 43 days. The average time from identification of the IPN to diagnosis of lung cancer in the TIR cohort was 177 days. There was a 134-day difference between the 2 groups from identification of an IPN to diagnosis of lung cancer.

Conclusion: Implementation of a formal IPN program can reduce time-to-diagnosis of lung cancer in patients presenting with an IPN by 134 days compared with traditional referral patterns. Formal IPN programs can result in a shift towards an earlier stage of diagnosis for lung cancer.

Key Words: bronchoscopic lung volume reduction, revision bronchoscopy

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Studies involving human subjects (including research on identifiable human material and data) must have been performed with the appropriate participants' informed consent in compliance with the Helsinki Declaration. As this was a quality improvement study, the local regulatory body at Inova Fairfax Hospital did not require ethical standard approval. This study protocol was reviewed and the need for approval was waived by the Inova Health System Institutional Review Board. Despite lack of need, all ethical standards were followed in this study.

All authors contributed equally to the production of this manuscript.

All data generated or analyzed during this study are included in this article. Further enquiries can be directed to the corresponding author.

Access to data can be requested by contact the corresponding author.

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Lung cancer is the leading cause of cancer-related death in the United States.¹ Five-year survival for patients diagnosed with lung cancer is 28.1%.² The majority lung cancers are identified incidentally and are diagnosed at an advanced stage.^{3,4} Early detection, biopsy, and treatment offer the best chance for improved survival in patients with lung cancer.⁵⁻⁷

Incidental pulmonary nodules (IPN) are nodules identified on imaging performed for a reason unrelated to the detection of lung cancer. Approximately 1.6 million IPNs are detected in the United States every year.⁸ Many IPNs are found on computer tomography (CT) scans in acute care settings, such as an emergency room, but only 30% of these nodules are followed up appropriately.⁹ While the majority of IPNs are benign, 1% to 2% are malignant.¹⁰ Failure to intervene on high-risk IPNs can result in delays in diagnosis and treatment.

IPN management programs in the United States have increased in number over the past decade. Lung navigators and specialized detection software are key components of successful IPN programs to streamline nodule triage and patient communication. Furthermore, IPN programs can be linked to nodule clinics where specialized physicians, such as

interventional pulmonologists and thoracic surgeons, see patients on multidisciplinary visits to expedite care.

This study analyzes the time-to-diagnosis of patients with an IPN diagnosed as lung cancer through a formal IPN program compared with a matched group of lung cancer patients diagnosed with lung cancer using traditional IPN referral pathways. Patients managed through traditional referral patterns often experience prolonged diagnosis pathways, see physicians unable to perform nodule biopsies or staging, and suffer from unnecessary visits. By streamlining IPN triage and workup, nodule biopsies can be expedited and time-to-diagnosis can be reduced. Reduction in time-to-diagnosis may translate into earlier cancer identification and potentially improved survival.

METHOD

This is a retrospective, single-center analysis over a 5-year period with the primary endpoint being the time-to-diagnosis of lung cancer in patients presenting through a formal IPN program compared lung cancer diagnosed through traditional IPN referral pathways. Chart reviews were performed for both groups to determine the date of the initial CT scan showing the IPN to the date of the procedure diagnosing lung cancer. Data were collected and analyzed from the Inova Interventional Pulmonology Quality Database and the EPIC (Epic System Corporation, Verona, WI) electronic medical record (EMR). Patients in the formal IPN program (FIP) pathway had their IPN identified using specialized radiology-based software and were contacted by a lung navigator to expedite follow-up at a dedicated IPN clinic. These IPNs were diagnosed as lung cancer. Patients in the FIP group initially presented through the health system emergency departments or hospital-based acute care facility. Patients were excluded if they had known primary lung cancer or known metastatic cancer to the lungs. Patients diagnosed with nodules through lung cancer screening were also excluded. A multidisciplinary tumor board, including interventional pulmonologists, medical oncologists, radiation oncologists, and thoracic surgeons reviewed all diagnosed lung cancers to determine an appropriate treatment plan.

A second cohort of patients was collected who were also diagnosed with lung cancer after presenting as an IPN. This cohort was referred to the IPN clinic through traditional referral (TIR) pathways. Patients included in the TIR group presented to the IPN clinic outside of the formal IPN pathway algorithm. They were not identified through advanced IPN software and did not have their care facilitated by an IPN navigator. The TIR group was matched to the FIP group by including an equal number of patients who underwent curative surgical resection for early-stage lung cancer and an equal number of patients who required medical therapy for advanced-stage lung cancer following identification of an IPN. This group was not matched for age, sex, race, smoking status, or

comorbidities. The TIR patients presented to the same hospital system over a 7-year period. In the 7-year period, 5 years occurred before implementation of the formal IPN program by the health system and 2 years overlapped with the start of the IPN program. Patients in this cohort eventually presented to the IPN clinic through traditional IPN referral pathways and were diagnosed with lung cancer by the same clinical decision-making as the FIP cohort. Demographic data for each group are described in Table 1. The patients in the TIR group did not have IPNs identified using specialized radiology-based software and did not have lung navigators contact them to expedite care to an IPN clinic.

The Medtronic LungGPS software (Medtronic Corporation, Minneapolis, MN) and an institutionally developed EHR-based nodule identification software program were used to identify IPNs in the FIP cohort. These software platforms utilized natural language processing (NLP) to flag keywords in radiology reports. Lung navigators reviewed all flagged reports after identification to confirm the diagnosis of an IPN. Patients whose IPNs were suspicious for lung cancer were contacted via telephone by lung navigators to expedite an appointment in the dedicated IPN clinic. If a biopsy was indicated, it was performed using advanced bronchoscopy methods (robotic and nonrobotic platforms), trans-thoracic needle aspiration (TTNA), or surgical biopsy. Pathologic or clinical staging was performed on all patients to assess for metastatic disease.

The TIR cohort had IPNs identified by imaging at outpatient imaging centers or acute care facilities not included in the health system’s formal IPN network. Imaging scans were ordered predominantly by practitioners outside of the health system network. There were no alerts to the ordering physician or the patient that IPNs were identified in a radiology report. No designated physician was responsible for referring the patient a specialty service for biopsy. Patients were eventually referred to the lung nodule clinic after consultations by multiple specialties such as general pulmonology, oncology, thoracic surgery, and interventional pulmonology before arriving to the IPN clinic. Following diagnosis of lung cancer with pathologic or clinical staging, these patients were also reviewed by the same multidisciplinary tumor board as the FIP group to determine a treatment plan.

RESULTS

Between May 2019 and May 2024, a total of 9303 patients were identified with IPNs through the formal IPN program. Ninety-four of these patients (1%) presented with IPNs that were biopsy-proven cancers. Eighty patients were diagnosed and staged using a combination of robotic and nonrobotic navigational bronchoscopy. These patients were concurrently staged using endobronchial ultrasound bronchoscopy (EBUS). Thirteen patients were diagnosed with CT-guided needle biopsy, and one by surgical biopsy. These

TABLE 1. Patient Demographic Data

	Formal IPN program (FIP) cohort	Traditional IPN referral pathway (TIR) cohort
Age (mean) ± IQR	72.0 ± 18	74.3 ± 18
Male:female	39:40	37:42
Active or former smoker (%)	53 (68)	60 (76)
TNM stage 1 or 2 NSCLC (%)	21 (27)	21 (27)
TNM stage 3 or 4 NSCLC (%)	58 (73)	58 (73)

14 patients were staged using positron emission tomography (PET) scans. Eleven of the 94 cancerous IPNs (12%) did not originate from the lung. These 11 IPNs were metastatic lesions to the lung. Eighty-three cancers originated from the lung. Seventy-nine of the 83 cancers were greater than 8 millimeters in diameter (Table 2). Of the 83 cancers originating from the lung, 4 patients (4%) were diagnosed with low-grade, typical carcinoid tumors. These patients were identified as having cancers originating from the lung and were identified in 5 types of radiographic studies (Table 3).

The 79 remaining patients (84%) were diagnosed with primary, non-small cell lung cancer (NSCLC) and made up the FIP group. Twenty-one of the 79 patients in the FIP group (27%) were diagnosed at an early stage and underwent curative surgical resection. All 4 of the patients diagnosed with low-grade, typical carcinoid tumors (100%) underwent surgical resection. Fifty-eight of the 79 patients in the FIP group (73%) were diagnosed with metastatic lung cancer at the time of biopsy and were referred to medical oncology for treatment (Fig. 1).

The patients making up the FIP group were compared with the TIR group. The TIR group was made up of 79 patients who were also diagnosed with lung cancer first appearing as an IPN, but followed traditional referral pathways. The average time from identification of an IPN to diagnosis of lung cancer in the FIP cohort was 43 days. The average number of days from identification of an IPN to the first IPN clinic visit in the FIP group was 20 days. The average number of days from first visit to the IPN clinic to diagnostic biopsy was 23 days in the FIP group. The average time from identification of the IPN to diagnosis of lung cancer in the TIR cohort was 177 days. The average number of days from identification of an IPN to the first IPN clinic visit in the TIR group was 159 days. The average number of days from first visit to the IPN clinic to diagnostic biopsy was 18 days in the TIR group. The difference in time-to-diagnosis of lung cancer between the 2 groups was 134 days.

DISCUSSION

While formal IPN programs are effective in identifying nodules and facilitating the diagnosis of lung cancer, the true clinical impact has yet to be evaluated. This study describes the value of a formal IPN program in reducing the time-to-diagnosis of lung cancer compared with traditional referral pathways. The shortened interval to diagnosis may result in earlier treatment and potentially improved survival.¹¹

This study describes a 134-day reduction in time-to-diagnosis of lung cancer with implementation of a formal IPN program. Unfortunately, delays in time-to-diagnosis of malignant IPNs are common when formal IPN programs are not available. Studies describe delays for lung cancer patients in presenting to a family doctor (median 21 d) and referral to a lung specialist (median 56 d) for nodule evaluation. In addition, delays occur for subsequent diagnostic tests (median 33 d) to establish the presence of

TABLE 2. Size of Nodules Diagnosed as Cancers Originating From the Lung

Nodule size	Malignant nodules originating from the lung
6-8 millimeters	4
> 8 millimeters	79

TABLE 3. Different Types of Imaging Studies Where Cancerous IPNs Were First Identified

Type of imaging study	No. cancerous IPNs identified
Chest-CT	41
Abdominal-CT	16
Angiogram of the Chest	20
Neck-CT	2
Chest X-Ray	4

lung cancer. Furthermore, the median time from initial symptoms of cancer until the start of treatment can be close to 6 months (189 d).¹² These delays in diagnoses and treatments of lung cancer can increase the risk for cancer progression and potentially worse outcomes.¹³ Delays in presentation from IPN identification to IPN clinic visits and clinic visits to diagnostic biopsy were also present in the FIP group. These were typically related to patient preference in scheduling appointments and availability of procedure space to perform biopsies.

Referral to a physician who performs nodule biopsies and mediastinal staging can minimize delays in lung cancer diagnoses. When IPNs are first identified, primary care physicians are often responsible for coordinating initial referrals to establish a diagnosis.¹⁴ Unfortunately, these traditional referral pathways can result in prolonged waiting times and unnecessary physicians visits. For example, primary care physician may refer a patient with an IPN to an oncologist, who then refers to a thoracic surgeon, who then refers to an interventional pulmonologist or interventional radiologist, who finally performs a biopsy.¹⁵ Furthermore, if a CT-guided needle biopsy is performed and pathologic mediastinal staging is necessary, additional time may pass before the patient undergoes an endobronchial ultrasound bronchoscopy (EBUS) (Fig. 2A). These multiple, unnecessary referrals can prolong time-to-diagnosis and can potentially increase the risk of lung cancer metastasizing. This pattern is common for IPNs identified outside of formal IPN programs and served as the most significant reason for the time-to-diagnosis of 177 days in the TIR cohort. Conversely, patients in the FIP cohort were directly referred to an IPN clinic for biopsy by dedicated lung navigators. As a result, patients in the FIP group were expeditiously referred, diagnosed, and staged for lung cancer with minimal delays (Fig. 2B).

Lung cancers diagnosed in the FIP group benefited from specialized computer software to identify IPNs. This software was based on identification of IPNs through natural language processing (NLP) and keyword searches performed on radiology reports.¹⁶⁻¹⁸ When the words “lung” and “nodule” were found in radiology reports performed on patients in the health system acute care facilities, lung navigators were alerted. NLP software requires humans to review and confirm findings to assess false positive results. More contemporary IPN software uses artificial intelligence (AI) and computational linguistics to analyze radiology reports to identify lung nodules.^{19,20} These advancements in identification software may improve sensitivity and specificity, resulting in fewer false positive findings.

IPN programs that integrate lung navigators to communicate with patients dramatically improve patient follow-up and expedite care.⁹ Traditionally, only 30% of IPNs identified outside of formal programs receive

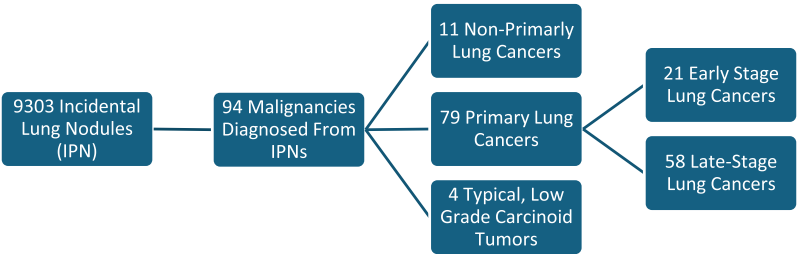


FIGURE 1. Flowchart for patients diagnosed with cancer through formal incidental nodule program.

appropriate follow-up.⁸ Failure to follow-up commonly stems from gaps in communication between physicians who order an imaging scan, primary care physicians, and patients. Previously published data from our institution described an 87% incidence of follow-up for patients identified with an IPN program with dedicated lung navigators compared with the previously cited 30%.¹⁰ Patients found to have high-risk IPNs through our program were immediately contacted by lung navigators to arrange for consultation with the IPN clinic.²¹ In fact, some of these clinic visits were arranged on the same day that an IPN was identified in the emergency department. “Fast-tracking” patients with IPNs to earlier clinic visits allowed for earlier diagnostic procedures and shorter time-to-diagnoses. Like IPN software, lung navigator positions can often be costly and may be difficult to justify when a formal IPN program is in initial development. In many instances, existing staff can

initially serve as lung navigators until additional personnel can be justified.

While formal IPN programs can reduce the time-to-diagnosis for lung cancer, the cost and implementation of IPN software can be significant. IPN software often requires a capital investment of over one hundred thousand dollars and integration into a medical system can be time-consuming. Furthermore, IPN navigators are essential for utilizing and managing this software. Due to the workload carried by IPN navigators, additional full-time equivalent (FTE) requests are typically necessary to employ enough personnel to manage the large number of patients evaluated by an IPN program. Unfortunately, these costs are commonly prohibitive to starting a formal IPN program. Fortunately, there are interventions that hospitals can perform to begin formal IPN programs. Institutions can also create IPN care pathways between interventional

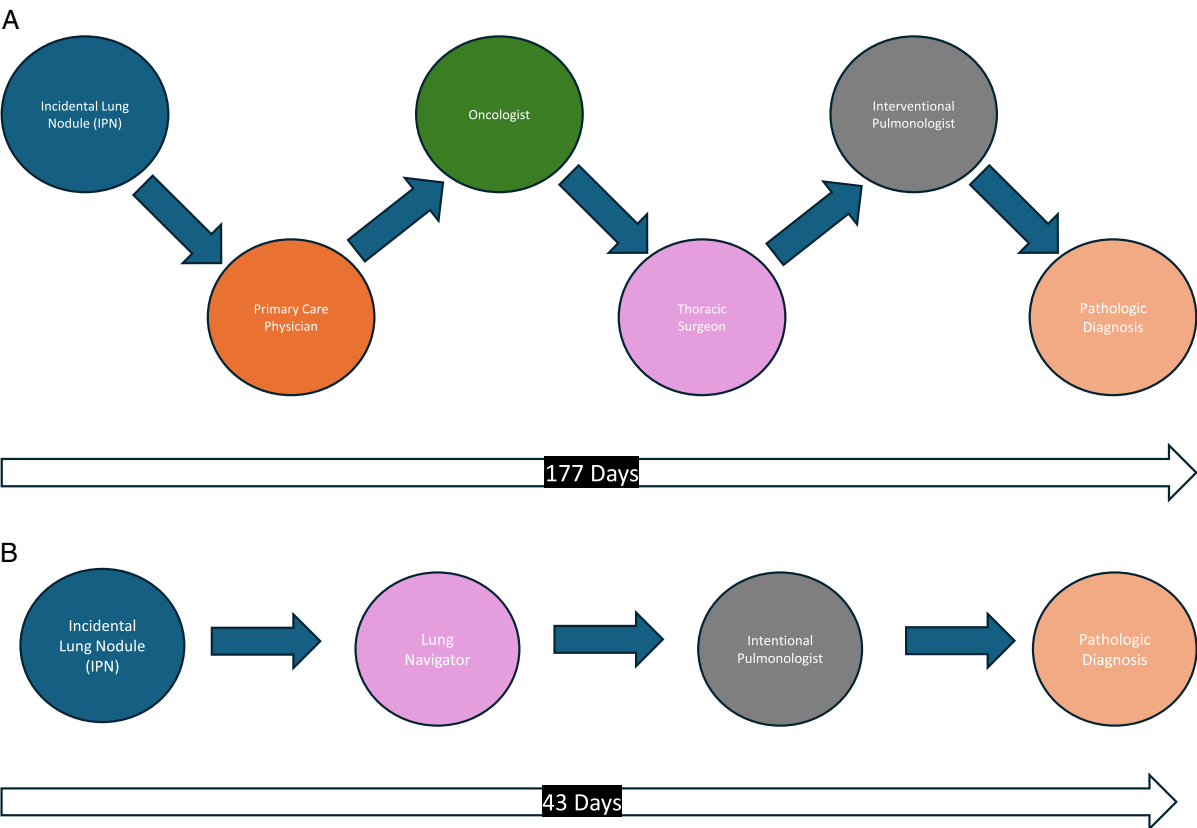


FIGURE 2. A, Example of traditional referral patterns for incidental lung nodules. B, Example of formal program referral patterns for incidental lung nodules.

pulmonologists and radiologists when an incidental lung nodule is identified. In addition, computer-based algorithms and pathways can be created through the hospital's existing EMR.¹⁰

This study shows that formal IPN programs can provide a stage shift for early lung cancer diagnoses.²² Nationwide, only 16% of lung cancers are diagnosed at an early stage, when surgical or radiation treatment is curative in intent.²² Conversely, the remaining 84% are diagnosed with metastatic disease, when cure is less likely. In the FIP cohort, 27% of patients with lung cancer underwent curative surgery after first presenting with an IPN. This is an 11% improvement in early-stage diagnosis and signifies a dramatic stage shift. More patients were diagnosed with lung cancer earlier and faster compared with traditional referral pathways seen in the TIR cohort. The implementation of formal IPN programs could result in thousands of patients being diagnosed earlier, resulting in improved survival.

The limitations of our analysis include the single-center, retrospective nature of the study. Single-center studies are prone to institutional biases. In addition, retrospective studies may be biased by incomplete data collection. While the absence of survival rates, stage-specific treatment outcomes, or cost-effectiveness metrics prevent appreciation of the large-scale impact of formal IPN programs, such outcomes will hopefully be the topic of future studies.

Formal IPN programs offer substantial benefits for patients with IPNs. These programs often have specialized software to streamline nodule identification and lung navigators to improve communication with patients. These interventions allow for shorter time-to-diagnoses of lung cancer and result in a stage shift towards early-stage diagnoses. While formal IPN programs often require capital investment for both equipment and personnel, the benefits are worthwhile. The improvements in time-to-diagnosis, stage shift, and potentially survival justify the widespread adoption of formal IPN programs.

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